

Chapter 10

Money, the Federal Reserve, and the Interest Rate

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What is Money?

- How is it that a store owner is willing to part with a steak and a loaf of bread that I can eat in exchange for access to some pieces of paper that are *intrinsically worthless*?
- Money is a collection of faith systems that functions as
 - a medium of exchange,
 - a store of value, and
 - a unit of account.
- Anything satisfying the three conditions can be identified as money.

A Medium of Exchange

- Economy without money is called an exchange economy, or a barter system.
 - A barter system requires a *double coincidence of wants* for trade to take place.
 - Money neatly eliminates the double-coincidence-of-wants problem, and makes the trade process efficient and simple.
- To satisfy the medium of exchange criteria, money should be portable enough to carry around easily.
 - huge wheels of carved stone vs. paper bills

A Store of Value

- Money should be able to transfer purchasing power from one time period to another.
- Alternative assets to store value
 - antique paintings
 - diamonds, golds,
- Advantages of money as a store of value
 - Liquidity: the efficiency or ease with which an asset or security can be converted into ready cash without affecting its market price.
 - Durability

A Unit of Account

- Money should also be able to serve as a unit of account—the standard of pricing and evaluation.
- A Nike running shoes is quoted as costing \$100, not 20 cups of coffee.
- Advantages of money as a unit of account
 - Consistency
 - Divisibility: dollar, quarter, cent

History of Money

- Various kinds of money in the history
 - sea shells: primitive economy
 - candy bars and cigarettes: World War II prisoner-of-war camps
 - huge wheels of carved stone: island of Yap in the South Pacific
 - beads: North American Indians
 - cattle: southern Africa
 - small green scraps of paper: contemporary North America
 - bird feathers: between the island of Santa Cruz and nearby Pacific Islands in the 19th century
 - dolphin teeth: Solomon Islands
- These various kinds of money are generally categorized into two groups: commodity monies and fiat money.

Commodity and Fiat Monies

- Commodity money: Items used as money that also have intrinsic value in some other use.
- Fiat money (token money): Items designated as money that are intrinsically worthless.
 - Currency debasement: legal tender is officially approved and socially agreed upon as money, but its value decreases when its supply is increased rapidly.

Are Cryptocurrencies Monies?

- Cryptocurrency is a currency that exists only in electronic form. It uses a technology called *blockchain* to maintain a public ledger that records transaction history.
- Bitcoin, the first of cryptocurrency, was introduced in 2009.
 - Bitcoins are neither commodity money nor fiat money.
 - There are proponents and opponents of bitcoins.
 - During bitcoin's brief history, its dollar value has fluctuated wildly.

Measuring the Supply of Money in the United States

- Recall that money is used to buy things (a means of payment), to hold wealth (a store of value), and to quote prices (a unit of account).
- These characteristics apply to a broad range of assets in the U.S. economy in addition to dollar bills.
- The most common measures of money are M1 and M2
 - M1: transactions money

$$M1 \equiv \text{currency held outside banks} + \text{demand deposits} + \text{traveler's checks} \\ + \text{other checkable deposits}$$

- M2: broad money

$$M2 \equiv M1 + \text{Savings accounts} + \text{Money market accounts} + \text{Other near monies}$$

How Banks *Create* Money

Goldsmith: the origins of the modern banking

- In the 15th and 16th centuries, gold was used as money, particularly for large transactions.
 - Gold is inconvenient to carry around and susceptible to theft, people began to place their gold with goldsmiths for safekeeping.
 - Goldsmiths charged small fees for keeping gold, and depositors got receipts as confirmation of storage.
 - These receipts were considered "as good as gold" and became a form of paper money.
- Realizing people did not come often to withdraw gold and having a large stock of gold on hand, goldsmiths started to *lend out* some of this gold.
 - The goldsmiths changed from mere depositories for gold into financial institutions that had the power to *create* money.

What is Money Creation?

- Note that goldsmiths' transformation from mere depositories to quasi-banks occurred as soon as they began making loans.
- The magic of loans lies in that the goldsmiths increased the amount of money (receipts in this case) in circulation by creating additional claims to gold without adding any more real gold to the system.
- In a nutshell, the key to the banking system lies in the credibility of the financial intermediaries and faith in the value of the money.
- Does it sound mysterious? Or, unsound or somehow dubious? Or, crystal clear and reasonable?

Bank Run (Bank Failure)

- Still, it is true that claims on gold were greater than the actual amount of gold in the economy.
- In normal times, the system worked well. However, once people began to worry about the goldsmiths' financial credibility and to have claims on deposits at the same time, the whole system would break down—we call this situation as bank run.
- A bank run scene from *It's a Wonderful Life* (1946)
- WSJ YouTube clip: How Silicon Valley Bank Collapsed in 36 Hours

The Modern Banking System

- The creation of money is simply an accounting procedure.
- A Brief review of accounting

$$\text{Assets} \equiv \text{Liabilities} + \text{Net Worth}$$

Understanding of Money Creation in Detail

- Required reserve ratio:

the percentage of its total deposits that a bank must keep as cash or reserves at the Federal Reserve

- A bank has deposits of \$100 and the required ratio is 20%.
How much is the required amount of reserves? How much is the excess amount of reserves?

actual reserves $\equiv$ excess reserves + required reserves

excess reserves $\equiv$ actual reserves – required reserves

- How much can the bank lend and still meet the reserve requirement?

A Single-Bank Economy

- For simplicity, assume that anyone who gets a loan keeps the entire proceeds in the bank and nothing is withdrawn as cash.

Panel 1		Panel 2		Panel 3	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
Reserves 0	0 Deposits	Reserves 100	100 Deposits	Reserves 100 Loans 400	500 Deposits

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▲ **FIGURE 10.2 Balance Sheets of a Bank in a Single-Bank Economy**

In panel 2, there is an initial deposit of \$100. In panel 3, the bank has made loans of \$400.

- With \$80 excess reserves, the bank can have up to \$400 of additional deposits.

The Money Multiplier

- Money multiplier: The multiple by which deposits can increase for every dollar increase in reserves.

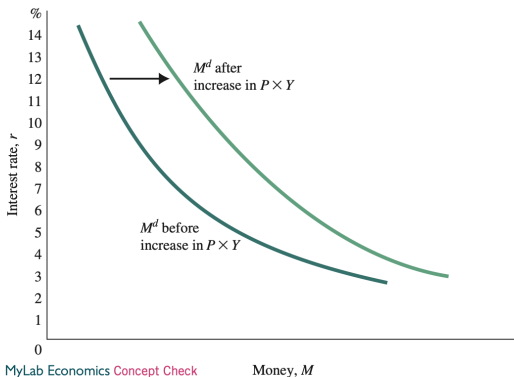
$$\text{money multiplier} \equiv \frac{1}{\text{required reserve ratio}}$$

Financial Portfolio of Household

- Think about the financial assets of a household.
 - Households hold financial assets, some of which are in the form of M1 money, which is more liquid but zero-interest-rate assets; some of which are in savings, bonds, and stocks, which are less liquid but yielding interest earnings.
 - Money demand refers to how many liquid assets people choose to hold in their financial portfolio.
- ! *Be careful not to confuse it with the desire for "money" itself.*

Determinants of Money Demand

- The quantity of money demanded depends on
 - interest rates: opportunity cost of holding money, rates of return when you hold financial securities instead of holding money
 - size of transactions: $P \times Y$



Interest Rates and Security Prices

- Before we discuss how the Fed controls the interest rate, we need to look at the relationship between interest rates and security prices.
- Take an example of an 10-year U.S. Treasury security (a government bond).
 - It promises to give you at maturity face value \$1,000 and to pay a coupon of \$20 per year during 10 years. If the equilibrium price of the bond is \$1,100, what is the interest rate? What is the interest rate when the bond price \$900?
 - $i_1 \approx 0.9\%$, $i_2 \approx 3\%$.
- Interest rates and security prices are inversely related.

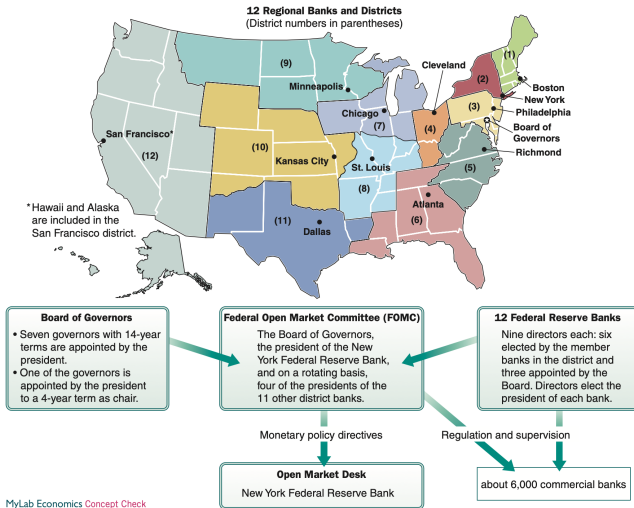
The Federal Reserve System

- Founded in 1913 by an act of Congress, the Fed is the central bank of the United States.
 - Bankers' bank: only banks (and occasionally foreign governments) can have accounts in them.
- The Fed is responsible for monetary policy in the United States.
- One of the important responsibilities of the Fed is to act as the *lender of last resort* for banking system.

The Organization of the Federal Reserve System

- The United States is divided into 12 Federal Reserve districts, each with its own Federal Reserve bank.
 - The district banks are like branch office of the Fed: they carry out the rules, regulations, and functions of the central system in their districts and report to the Board of Governors on local economic conditions.
 - The Board of Governors is the most important group within the Federal Reserve System: it consists of seven members, each appointed 14 years by the president of the United States.

The Structure of the Federal Reserve System



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▲ **FIGURE 10.4** The Structure of the Federal Reserve System

How the Fed Controls the Interest Rate

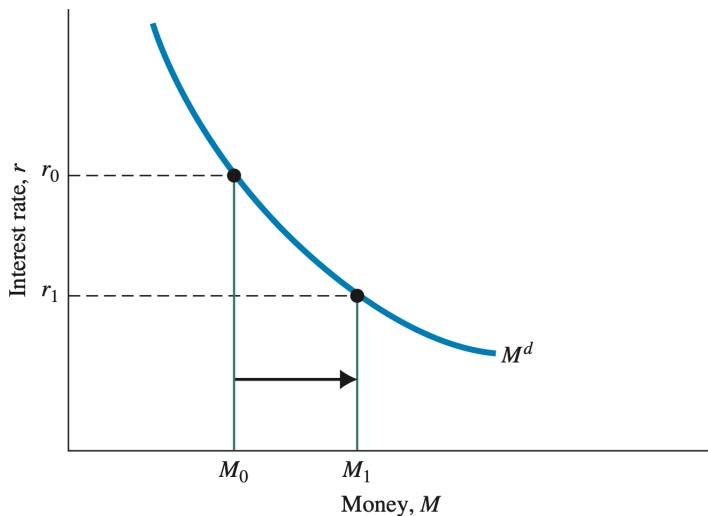
- U.S. monetary policy is formally set by the **Federal Open Market Committee (FOMC)**.
- The FOMC sets goals concerning interest rates, and it directs the **Open Market Desk** in the New York Federal Reserve Bank to buy and/or sell government securities (open market operations).

How the Fed Controls the Interest Rate

Conventional monetary policy

- Tools Prior to 2008 (conventional monetary policy)
 - The Fed had three tools available to control the interest rate by changing the money supply: open market operations, changing the reserve requirement ratio, and changing the discount rate that banks pay to the Fed to borrow reserves.
- The Fed can set the value of the money supply, the quantity of money.
 - Given the quantity of money that the Fed chooses to supply, the interest rate can simply be read off the money demand schedule.

The Equilibrium Interest Rate



Global Financial Crisis

- Beginning in about 2003, the U.S. economy experienced a "housing bubble".
 - Financial institutions issued mortgages with less oversight, in some cases to households with poor credit ratings—subprime borrowers.
 - When housing prices began to fall in late 2005, the stage was set for a financial crisis.
- In March 2008, the Fed began to broaden its role in the banking system to respond to the crisis.
 - The Fed attempted to bailout Bear Sterns by guaranteeing \$30 billion of its liabilities to JPMorgan.
 - The Fed participated in a government takeover of Fannie Mae and Freddie Mac.
 - The Fed loaned \$85 billion to the AIG insurance company to help it to avoid bankruptcy.
 - The Fed urged Congress to pass a \$700 billion bailout bill.

Tools After 2008

Unconventional monetary policy

- When the economy hits the zero lower bound of interest rate, conventional monetary policy is not effective anymore.
- Quantitative easing
 - a type of monetary policy by which a nation's central bank tries to increase the liquidity in its financial system
 - more aggressive involvement in financial markets
- Forward guidance
 - a tool that central banks use to provide communication to the public about the likely future course of monetary policy.
 - When central banks provide forward guidance, individuals and businesses will use this information in making decisions about spending and investments. Thus, forward guidance about future policy can influence financial and economic conditions *today*. (expectation-formation and money demand changing tool)